

Panelists



Baburao Rane, Digital Connectivity and User Experience Specialist, TATA Motors



Bob Paul Raj, Head - Connected & Digital Platforms, Mahindra & Mahindra



Helge Sverre Eide, Partner and Managing Director, Wireless Power and Communication



Imran Mulla, Head of R&D, Classic Legends (JAWA Motorcycles, Yezdi, BSA)

grow crops. An example of precision farming is spraying pesticides in the affected areas with the help of drones. Here, drones are utilised to scan the field and identify pest affected areas using image processing techniques. Once the areas are identified, pesticides can be sprayed on those areas. With precision farming, time is saved, yield is maintained, cost of the pesticides reduces, and the overall productivity increases, thus helping us achieve economies of scale.

Bob adds, connectivity must be adaptable to all kinds of machinery. Decisions must be made by OEMs on how much data needs to be on cloud, and how much should be processed on the edge. Once we have access to all the data collected, meaningful insights can be generated. He says, "In agriculture, we have people who do processing of agri-inputs using the data like quantity of produce, delivery data, and quality of produce." Additionally, ensuring a secure way for this data to transfer across different systems, not only on the cloud level but also within the system itself, is also essential.

Need for a connected approach

Adoption of a new technology in any industry relies most importantly on collaborations. The data harvested from sensors provides a lot of insights to the users and OEM. Rane thinks that the next big challenge to all the OEMs will be trying to see how to monetise this data, possibly with partners who are interested in this data for improving technologies even further.

The challenges encountered by all OEMs will more or less be the same, so there can be a common neutral

platform where data can be shared and then allow third parties or start-ups to engage. This, he believes, will be a good way for this ecosystem to accelerate much faster.

Helge Sverre Eide, Partner and Managing Director, Wireless Power and Communication, adds, "I think it's a global opportunity for growth and it's a big opportunity to move things into a safer and more sustainable position. We need to agree on the data format, and we need to be able to manage the cyber security aspects and the individual person's privacy. In doing that, cooperation is critical."

Greater data needs greater security

Earlier, the vehicle data passed through the CAN networks inside the closed ecosystem of the vehicle. The journey for connected mobility has just begun, and basic telemetry features are already being implemented like basic track and trace. But with the increasing demand for connected mobility, data has to be exposed to an open cloud, and in such events there is always a possibility of theft of data, data manipulation and hacking.

Rane says, "Cybersecurity would be a key element. Because earlier it was a closed loop. Data was coming in, and was getting stored inside the vehicle itself, without any external interference. But now it is being exposed to an open network. And without proper security, anyone can hack or maneuver the data, causing serious accidents."

There are various smart devices from different vendors integrated inside an electric vehicle. According to Rane, the privacy of data is compro-

mised when third-party services get involved and expect some data from the vehicles.

Eide warns that if a sub-vendor wants to access the data from a particular part of the vehicle, it is important that other data remains unaffected and protected, for which segregation of data is necessary.

The road ahead

The idea of connected mobility has gained much traction in the past couple of years but there is still minimal innovation and development in the field.

Imran believes this is because we are currently in the second stage of evolution. The first stage was to understand the challenges associated with this space and entering the space. He believes that the journey from here is going to be pretty quick.

In the last couple of years, each OEM has put specific focus on technologies, starting off with EVs, engine control units' advancements, and finally touched connected mobility. Imran says, "The journey we have done so far primarily involves growth in each of these areas. In the next five years or so, we will witness a very high growth as far as each of these areas is concerned. India has got the capacity and capability in terms of skill set. The intent is much clearer now, and the vision and the path has largely been made clear as well." **EFY**

The article is an extract from a panel discussion at the MOVES 2021.

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